02 Registered targets and complete differential cuts

A scalar target fixes the cotangent; each relevant path crosses the cut once

Native scalar units

Token × channel; final CLS channel

$$O = \mathrm{concat}_h(A_h V_h) W_O + b_O$$

Aggregated responses

Patch sum; activation-weighted energy

$$\mathrm{d}O_h = (\mathrm{d}A_h) V_h + A_h \mathrm{d}V_h$$

Q + K = routing; cotangent includes W_O

Structured paths

Heads; Q / K / V; identity / branch

$$y = u + f(u), \quad g_y = \nabla_y F$$

Output targets

Ground-truth and frozen clean-top1 logits

$$E_y = E_{\text{identity}} + E_{\text{branch}}$$